MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-M101/BSM101 Mathematics –IA

UPID : 001004

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) State whether the given system of equation have one, many or no solution:
$x - 3y + 5z + 2 = 0$
$3x - 9y + 15z - 3 = 0$

(ii) Is Zero transformation a Linear Transformation? Give reason.

(iii) Is {(1, 1, 0), (0, 1, 1), (1, 0, 1)} basis of R^3 ? Justify.

(iv) Write the value of $\beta(1/2, 1/2)$.

(v) Is the function $f(x) = \log(x+1)$ obeys L.M.V.T. in the interval [1,5] ?

(vi) Every square matrix A can be expressed as a sum of symmetric and skew symmetric matrix and it can be expressed as=

(vii) A non-null vector is linearly_____.

(viii) What is eigen value of a square matrix? Explain with an example.

(ix) Define Beta and Gamma functions.

(x) If $f(x)=4x-x^2$ satisfies Rolle's theorem in [0, 4] then c=?

(xi) Prove that the product of any matrix and its transpose is a symmetric matrix.

(xii) If $T:V \to W$ is a Linear Transformation such that $\ker(T)=\{\theta\}$. If dimension of V is r, then find dimension T(V).

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Check whether the vectors (2, 6, -1, 8), (0, 10, 4, 3), (0, 0, -1, 4) and (0, 0, 0, 8) are linearly independent in four-dimensional vector space R^4 or not. Is it a basis ? Justify. [5]

3. Answer the following: [5]

Show that $W = \{(x, y, z) \in R^3 : x + y + z = 0, 2x + y - z = 0\}$ is a subspace of R^3. Find a basis of W. What is its dimension?

4. Answer the following: [5]

If possible diagonalise the matrix $\begin{pmatrix} 10 & 2 \\ 2 & 7 \end{pmatrix}$.

5. Answer the following: [5]

Evaluate $\int_0^{\frac{\pi}{2}} sin^4 x cos^5 x \, dx$

6. Expand sinx in a Maclaurin's series with nth remainder. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Answer the following: [8]

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The line segment $x + y = 1, 0 \leq y \leq 1$ is revolved about y-axis to generate a cone. Find the lateral surface area of the cone.

(b) Find the maxima and minima of the function $f(x) = x^5 - 5x^4 + 5x^3 - 1$ [7]

8. [15]

Prove that the mapping $T: R^3 \longrightarrow R^3$ defined by

$T(x, y, z) = (-2x + y, -x + 2y + 4z, 3x + z)$ is a linear transformation. Find the matrix of T in the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$, where $\alpha_1 = (-1, 2, 1), \alpha_2 = (2, 1, 1), \alpha_3 = (1, 0, 1)$.

9. (a) Answer the following: [8]

Find a Linear Transformation $T: R^3 \longrightarrow R^3$ whose kernel is generated by $(1, 2, 3, 4)$ and $(0, 1, 1, 1)$.

(b) Answer the following: [7]

Using Gauss-Jordan Elimination method solve the system of equation

$$x + y + z = 9$$
$$2x - 3y + 4z = 13$$
$$3x + 4y + 5z = 40$$

10. (a) Prove that $\int_0^\infty \frac{1}{(1+x^2)^3} dx = 35\pi/256$ [7]

(b) Obtain the evolute of the parabola $y^2 = 4ax$. [8]

11. Define Inner product space with an example. [15]

Use Gram-Schmidt process to convert the basis $\{(1, 2, -2), (2, 0. 1), (1, 1, 0)\}$ of R^3 into an orthogonal basis and next to an orthonormal basis.

*** END OF PAPER ***

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